

Valentinus Mahendra Aaron Quendangen

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https://ronaaron61.github.io/#/about

Motivated to work hard in carrying out work activities to achieve success and gain experience, such as learning new things to gain more experience, especially by doing or helping with a project or research.

Education

- **SMAK Santo Yoseph Denpasar** 2016 - 2019
  Science Major
- **Universitas Airlangga** 2019 - 2023
  Biomedical Engineering – Medical Instrumentation Major
  GPA/IPK: 3.68

Experiences

Research Assistant Project Jakarta
Medical Technology Cluster - Indonesian Medical Education and Research Institute (IMERI) FKUI May 2026 - Present

- Developed and submitted research proposals, including budgets, methodologies, and timelines
- Acted as a liaison between doctors, specialists, and external partners.
- Supervised interns, led site visits, and served as a speaker in academic sessions.
- Assisted in research data collection and verification.

IPSRS – Medical Equipment Maintenance Pontianak
Rumah Sakit Umum Medika Djaya November 2025 – May 2026

- Weekly and monthly checks on medical equipment in the ICU, ward, physiotherapy, emergency department, and the medical gas area.
- Conduct troubleshooting, calibration, and repair of faulty medical equipment.
- Support the installation of new equipment and conduct accompanying user training.
  - Maintenance, Problem Solving, Communication, Electricity, Soldering, Electromedic

ENT Product Specialist Surabaya
PT Edison Duta Sarana July 2024 – October 2024

- Manage the supply and stock of ENT medical devices and instruments.
- Communicating between principals, purchasing, marketing, and doctors about the products needed, including instruments, ENT Unit, and endoscopy camera.
- Training new users on using the product, assisting in the use of the medical equipment during surgery, and troubleshooting product problems.
- Planning and creating product lists, specifications, price quotes, and presentations
  - Speaking (Indonesian and English), Communication, ENT basic, Management, Problem Solving

Research Engineer Surabaya
PT Medika Karya Airlangga December 2023 – November 2025 (*Full Time*) – Present (*Part Time*)

- Research and development of a smart phototherapy system for jaundiced babies - Smart Phototherapy System Airlangga Bilirubin Nesting (AirBiliNest) Model 2 and 3. Including prototyping the electrical components, design, programming, application, and data gathering.
- Charged with managing technical issues, including prototyping, PCB, wiring design, and assembly of the controller, lamp design, sensor development, microcontroller programming, technical drawings, Android application development, and web development.

- Gather safety and efficacy data in Airlangga University Hospital, and process the data for the paper and future development.
    - Troubleshooting, basic electronics, soldering, Microcontroller, PCB design, Graphic Design, Laser Cutting, Soldering, Communication, Testing, Demonstrating, RnD, Android Studio, 3D Printing Design, front-end Web Design, back-end web design, Autodesk Inventor

Research Engineer Assistant Surabaya

Neonatal Research Group FK Universitas Airlangga August 2023 - Present

- Develop the research of Smart Phototherapy System Airlangga Bilirubin Nesting (AirBiliNest), which is a smart phototherapy device for jaundiced babies that can be monitored and controlled using an Android application. Assembling, testing, and multiplying the prototype required.
- Gather safety and efficacy data in Airlangga University Hospital, and process the data for the paper and future development.
- Implementing the MBKM Matching Fund events related to this device with the students, faculty, and industry partners in West Java.
    - Troubleshooting, Microcontroller, Laser Cutting, Soldering, Communication, Testing, Demonstrating

Merdeka Belajar Kampus Merdeka (MBKM) Matching Fund Surabaya

PT Integrasi Bisnis Eksekutif (IBE Reality) September 2022 - January 2023

- Assisted with the manufacturing of X-Mano, the Wearable Exoskeleton Hand Robot for Movement Therapy of Post-Stroke Patients, through the 4-month internship. Including the 3D design for the case, microcontroller, assembly, and Android application.
- Assisted the clinical trial to gain feedback from 20+ post-stroke patients in Universitas Airlangga Hospital.
    - 3D Printing, Autodesk Inventor, Microcontroller, Clinical trial, Entrepreneurship

Mentor on Coding Workshop for Biomedical Engineering Surabaya

Himpunan Mahasiswa Teknik Biomedis (HMTB) UNAIR March 2021 - June 2021

- Teach the basic use of syntax, functions, and how to make simple programs using C++ and Python for the Biomedical Engineering students class of 2021.
- Check and evaluate the results of the program creation task.
    - C++, Python, Teaching, Speaking

Technical Expertise

Programming (Python/MicroPython, C++/Arduino, Matlab, HTML) | Arduino, ESP, Analog Discovery | Basic electronics, Electrical circuits, analog and digital electronics, Soldering, PCB design, 3D Printing, Laser cutting, Graphic design, Autodesk Inventor/solidworks.

Achievements

- Best teamwork at Sebelas Maret International IoT Challenge 2021 – Smart Trash Bin to prevent COVID-19
- Silver award for iMIT SIC 2021 (4th International Malaysia-Indonesia-Thailand Symposium on Innovation and Creativity 2021) held by UiTM Cawangan Malaysia,
  “3d (Three-Dimensional) Lung Cancer Detection and Identification Based On Mixed Reality”

Research and Publications

- Edukasi Siswa Sma Negeri 1 Kampak Kabupaten Trenggalek Provinsi Jawa Timur Melalui Pelatihan Rancang Bangun Alat Deteksi Kelelahan Berbasis Audiovisual Dalam Peningkatan Mutu Kerja Dan Kesehatan. (Vol. 4 No. 2: Jurnal Pepadu, DOI:10.29303/pepadu.v4i2.2370) 2023
- Design Of Mulifrequency Electrical Impedance Tomography (Mfeit) Based On Analog Discovery To Detect Breast Cancer. (Vol. 87 No. 1: Jurnal Teknologi, DOI:10.11113/jurnalteknologi.v87.21921) 2024
- Evaluation of New Portable Phototherapy Device for Neonatal Hyperbilirubinemia (International Research Journal of Multidisciplinary Scope (IRJMS), DOI:10.47857/irjms.2025.v06i02.03766) 2025

Mini Projects

Projects: https://ronaaron61.github.io/#/projects

- Electrical Impedance Tomography using ESP Microcontroller to detect anomalies in a phantom that looks like a human body
- Bioimpedance measure using a Microcontroller, and can be used to detect hand angle estimation to control the mechanical arm
- Simple EMG and ECG module for a microcontroller using AD620
- Virtual mouse and keyboard that can be controlled by finger position in the webcam using Python and the mediapipe library
- A smart trash bin that can be opened by gently kicking the bin or placing a hand near the sensor. Can be connected to the Blynk app to monitor the trash volume